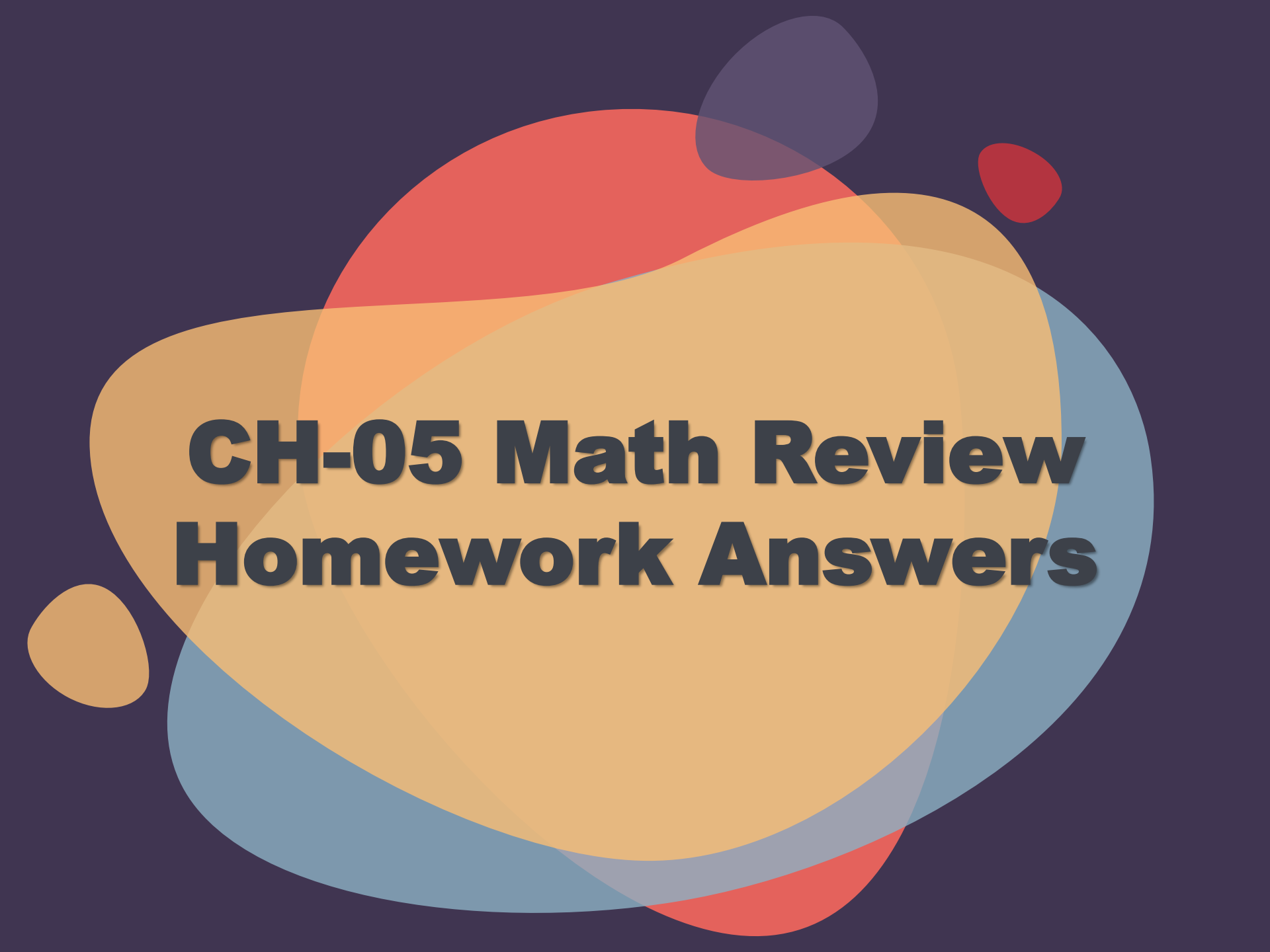
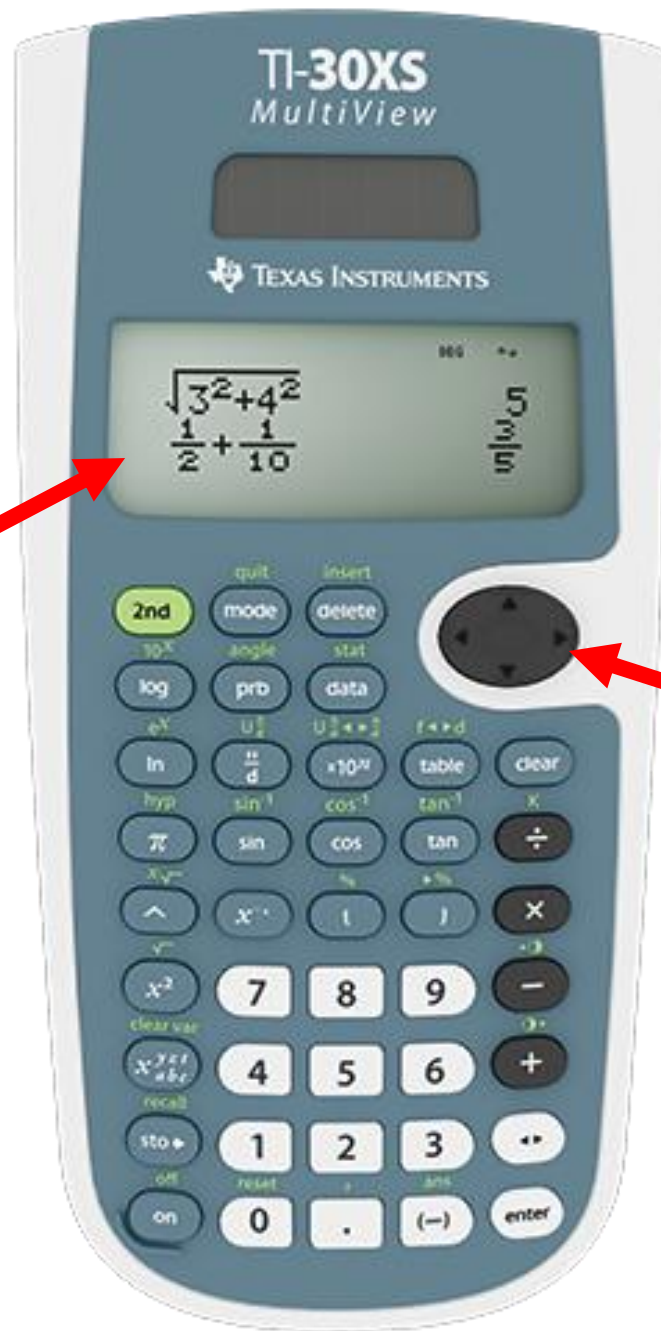




CH-05 Math Review Homework Answers





01

What is the airflow in an 18" diameter round duct with a velocity of 2500 fpm?

$$Q = VA$$

A. 2,940 cfm

B. 4,420 cfm

C. 35,300 cfm

D. 635,900 cfm

Step 1: Calculate the air velocity **$V = 2500 \text{ fpm}$**

Step 2: Calculate the cross-sectional area of the duct

$A = \pi r^2$, where D = diameter in feet (18 inches = 1.5 feet),
radius $r = 0.75 \text{ ft}$

$$A = (0.75)^2 \times 3.14 = \mathbf{1.767 \text{ ft}^2}$$

Step 3: Calculate airflow

$$Q = VA = (2500 \text{ ft/min}) (1.767 \text{ ft}^2) = \mathbf{4420 \text{ cfm}}$$



02

Solve $3+4\left(\frac{2+6}{2.5}\right)^{4.1}$

A. 474.17

B. 34461.64

C. 824.54

D. 635.74



03

Calculate $1000(1 + 0.043)^3 =$

A. 2165.723

B. 866.892

C. 1134.626

D. 1687.345



04

$$X = 5^2 5^3, \text{ find } X$$

A. 25

B. 125

C. 625

D. 3125



05

$$60 = 4^X, \text{ find } X$$

A. 2.95

B. 15.0

C. 19.1

D. 891

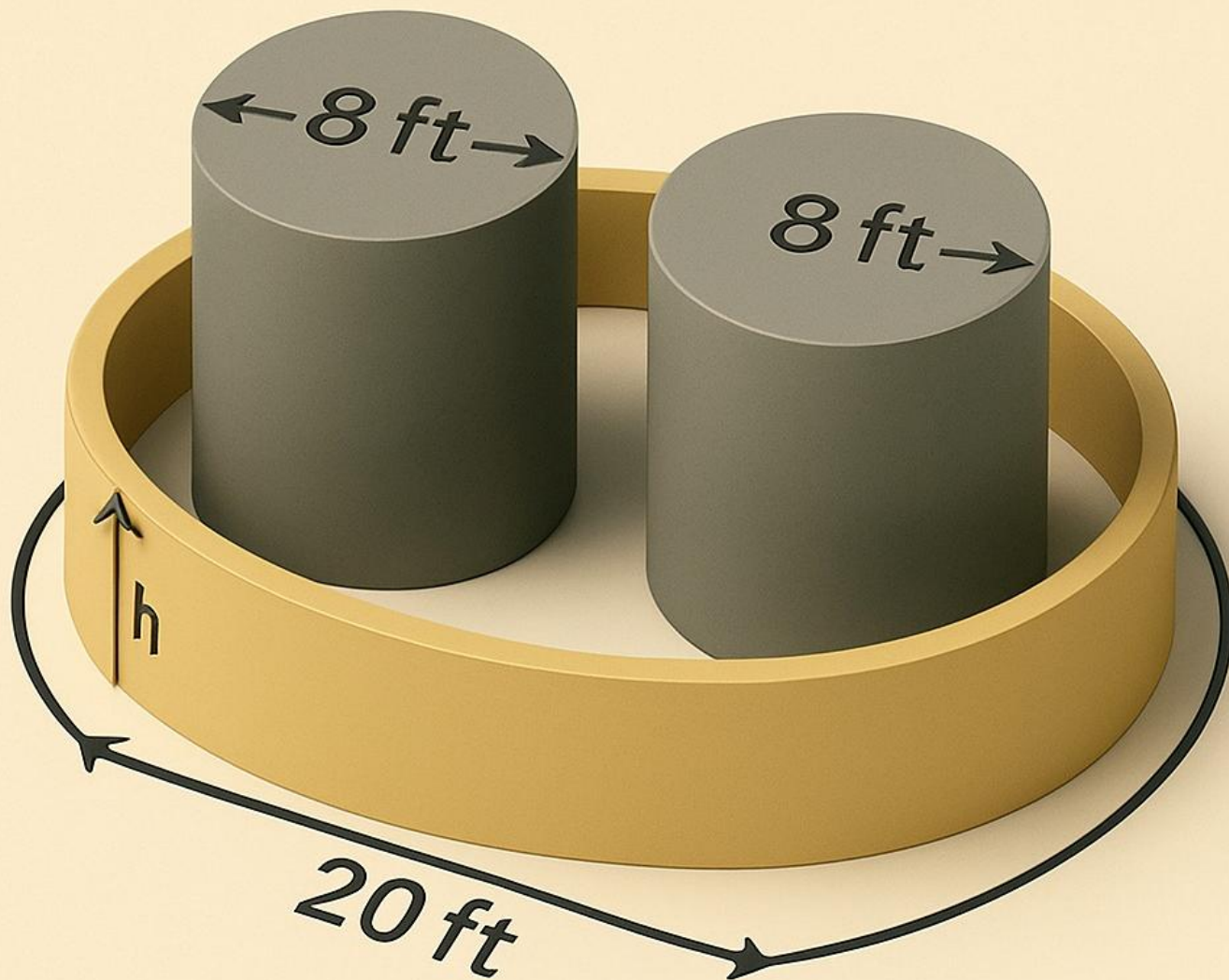


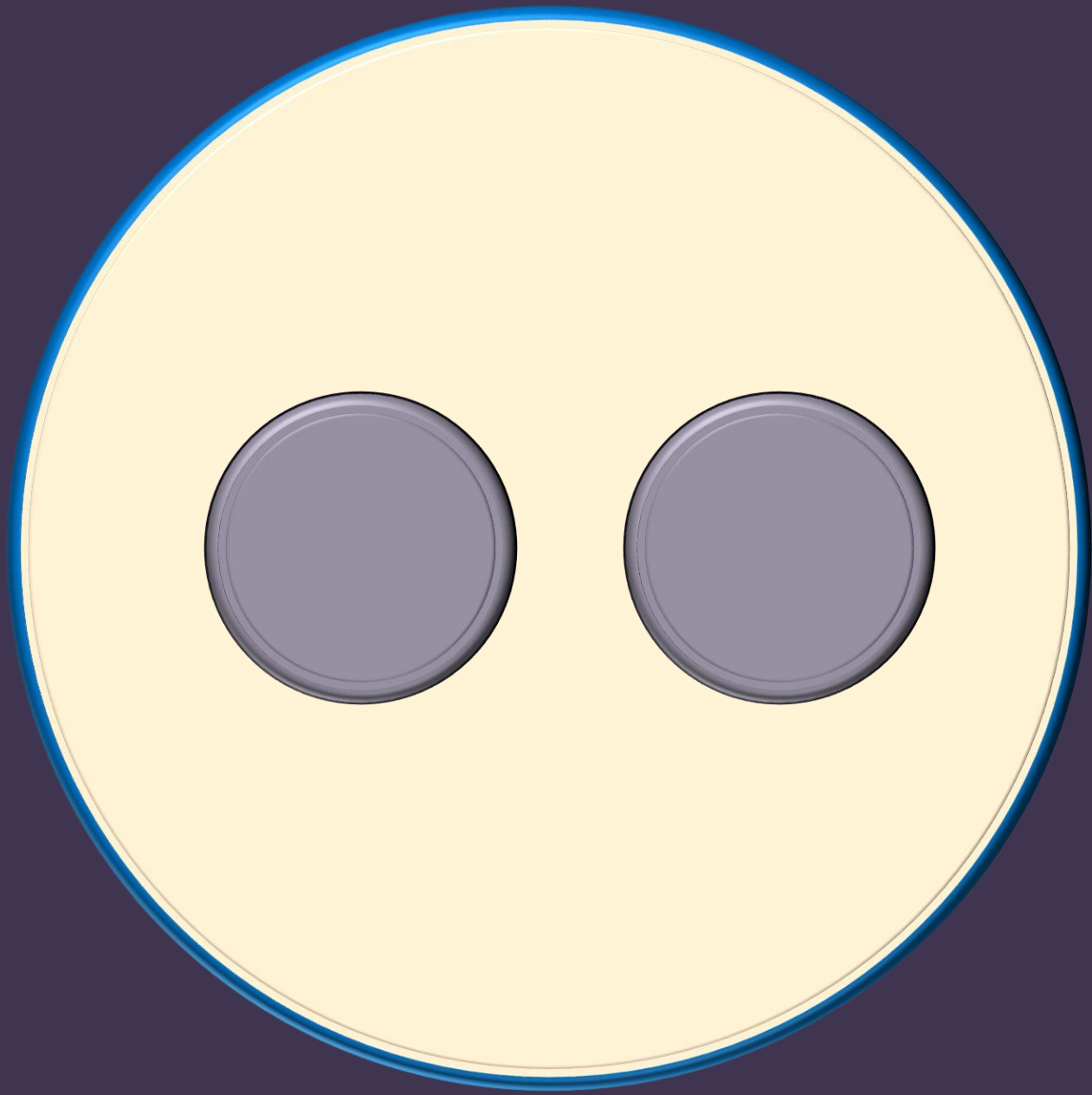
06

A spill containment wall is built around the perimeter of two 4,000-gallon tanks. The perimeter must contain the contents of one tank

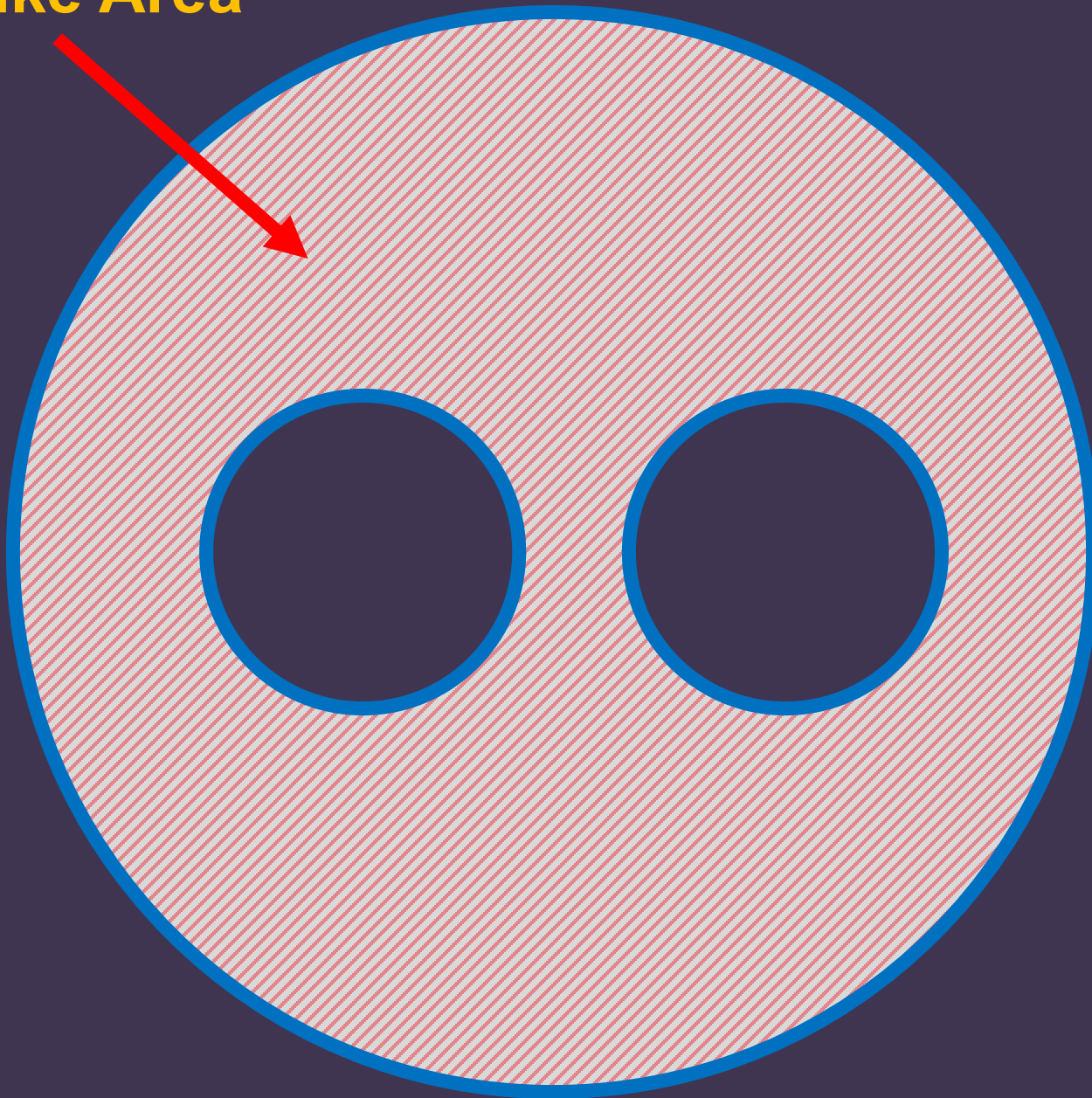
plus 10%. The footprint of each tank is an 8-foot circle (i.e. diameter). The containment wall forms a 20-foot circle (i.e. diameter). What is the minimum height the wall needs to be?

- A. 12 inches
- B. 18 inches
- C. 24 inches
- D. 30 inches





Active Dike Area



Step 1: Calculate Dike Volume (needed)

$$V = 4000 \times 1.1 = 4400 \text{ gallon} = \mathbf{588.2 \text{ ft}^3}$$

Step 2: Calculate Dike Total Area

$$A = \pi r^2, \text{ where } D = 20\text{foot}, \mathbf{r = 10 \text{ f}}$$

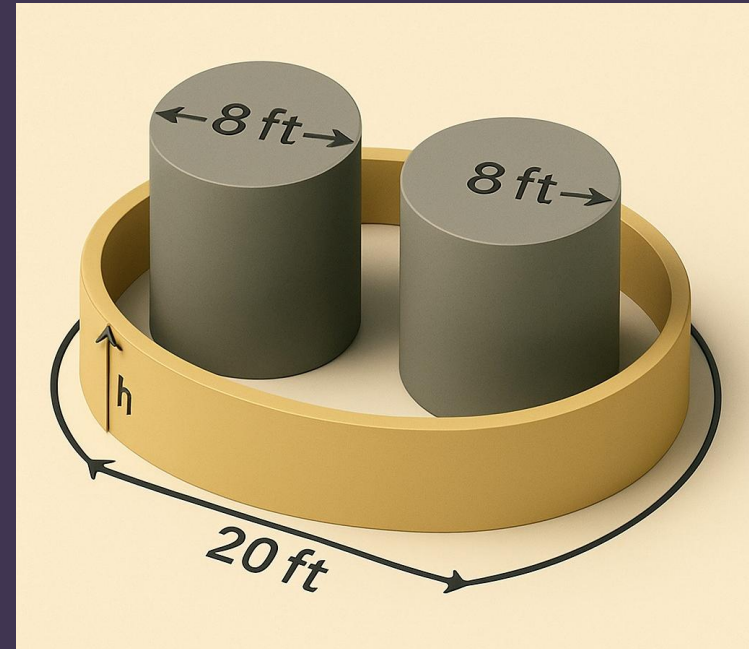
$$A = (10)^2 \times 3.14 = \mathbf{314.16 \text{ ft}^2}$$

Step 3: Calculate the total Tanks Area

$$A = \pi r^2, \text{ where } D = 8\text{foot}, \mathbf{r = 4 \text{ f}}$$

$$A = (4)^2 \times 3.14 = \mathbf{50.3 \text{ ft}^2}$$

$$\text{Total Tanks Area} = 50.2 \text{ ft}^2 \times 2 = \mathbf{100.6 \text{ ft}^2}$$



Step 4: Calculate the Active Dike Area

Active Dike Area = Total Dike Area – Total Area of the two Tanks
(footprint)

$$= 314.16 - 100.6 = 213.56 \text{ ft}^2$$

Step 5: Calculate Dike Height (*Volume = Area X Height*)

$$h = V/A = (588.2 \text{ ft}^3) \div (213.76 \text{ ft}^2) = 2.75 \text{ ft} \approx 32 \text{ inches}$$

A. 12 inches

B. 18 inches

C. 24 inches

D. 32 inches

